Project: Worked examples
Title: Section properties
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates area, centroid, second moments, moduli, radii of gyration, and torsion and warping constants of a parametric, polygon or composite section. The outline has x horizontal and y vertical; properties use EN 1993 axes, y-y horizontal and z-z vertical.

WHAT IT CALCULATES	RELATION
Area and centroid	$A = \tfrac{1}{2}\oint(x\,dy - y\,dx)$; $x_c = \int x\,dA / A$; $y_c = \int y\,dA / A$
Second moment about y-y	$I_y = \int y^2\,dA - A\,y_c^2$, by Green's theorem and the parallel-axis theorem
Second moment about z-z	$I_z = \int x^2\,dA - A\,x_c^2$; product $I_{yz} = \int xy\,dA - A\,x_c\,y_c$
Principal axes	$\tan 2\theta = -2\,I_{yz} / (I_y - I_z)$
Elastic section modulus	$W_{el,y} = I_y / y_{max}$
Plastic section modulus about the equal-area axis, found by bisection	$W_{pl,y} = \lvert Q_{above}\rvert + \lvert Q_{below}\rvert$
Torsion constant J	Saint-Venant series (solid rectangle), $\pi d^4/32$ (circle), Bredt (closed thin-walled), $\Sigma bt^3/3$ (open)
Warping constant of an I-section	$I_w = I_z\,(h - t_f)^2/4$

Basis: Properties of sections; torsion of solid and thin-walled sections[1](Table A.1 and Table 10.1, 8th ed.); Torsional properties of rolled I-shapes[2](App., 1st ed.); Hot rolled steel channels, I and H sections: dimensions and masses[3]; Hot finished structural hollow sections: tolerances, dimensions and sectional properties[4].

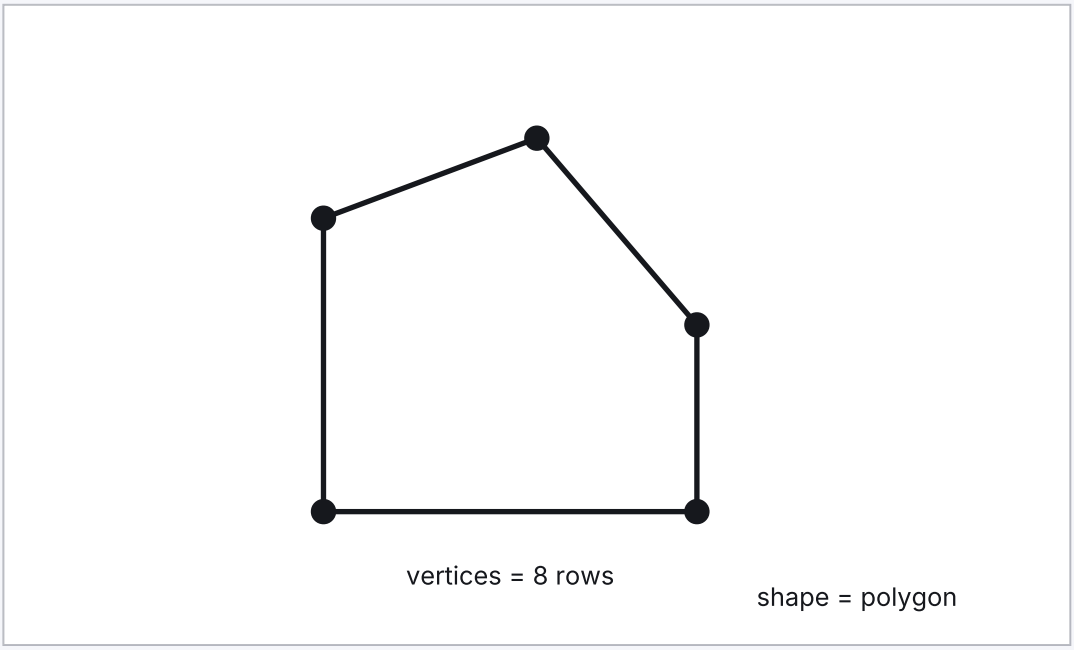


Figure 1 — A polygon through the vertices as listed (drawn representatively). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Shape	polygon		drawing: DRG-BG-100 rev C	any
vertices	Polygon vertices (polygon shape)	8 row(s)		drawing: DRG-BG-100 rev C	any

TABLE ROW	VALUES
vertices[1]	x = -200 mm; y = -300 mm; loop = 1
vertices[2]	x = 200 mm; y = -300 mm; loop = 1
vertices[3]	x = 200 mm; y = 300 mm; loop = 1
vertices[4]	x = -200 mm; y = 300 mm; loop = 1
vertices[5]	x = -180 mm; y = -280 mm; loop = 2
vertices[6]	x = 180 mm; y = -280 mm; loop = 2
vertices[7]	x = 180 mm; y = 280 mm; loop = 2
vertices[8]	x = -180 mm; y = 280 mm; loop = 2

3. Assumptions

A1. A polygon outline does not cross itself. (Green's theorem integrates the signed loops as given.)
A2. Holes lie inside the outline and do not overlap one another. (Green's theorem integrates the signed loops as given.)
A3. Rolled-section root fillets are quarter circles of radius r and flange tips are square. (Published section tables use the same idealisation.)
A4. Hollow-section corners are quarter circles of radius 1.5 t_w outside and 1.0 t_w inside by default. (BS EN 10210-2 hot-finished idealisation.)
A5. Thin-walled torsion constants neglect the junction and tip effects except the fillet term kept for rolled I-sections. (Open sections: $\Sigma bt^3/3$; closed: Bredt.)

4. Working

$[2]\ A = \tfrac{1}{2} \oint (x\,dy - y\,dx)$ [1](Table A.1)

$= \text{Green's theorem over 8 vertices in 2 loop(s)}$

$= 384\ \text{cm}^2$

Area

$[3]\ x_c = \int x\,dA \,/\, A$

$= \text{Green's theorem over 8 vertices in 2 loop(s)}$

$= 0\ \text{mm}$

Centroid x from the shape origin

$[4]\ y_c = \int y\,dA \,/\, A$

$= \text{Green's theorem over 8 vertices in 2 loop(s)}$

$= 0\ \text{mm}$

Centroid y from the shape origin

$[5]\ I_y = \int y^2\,dA - A\,y_c^2$

$= \text{Green's theorem over 8 vertices in 2 loop(s)}$

$= 193150\ \text{cm}^4$

Second moment of area about the centroidal y-y axis

$[6]\ I_z = \int x^2\,dA - A\,x_c^2$

$= \text{Green's theorem over 8 vertices in 2 loop(s)}$

$= 102270\ \text{cm}^4$

Second moment of area about the centroidal z-z axis

[7] $I_{yz} = \int xy\ dA - A\ x_c\ y_c$

= Green's theorem over 8 vertices in 2 loop(s)

= 0 cm⁴

Product of inertia about the centroidal y-y and z-z axes

[8] $I_1 = (I_y + I_z)/2 + \text{sqrt}(((I_y - I_z)/2)^2 + I_{yz}^2)$

= (193150 cm⁴ + 102270 cm⁴) / 2 + sqrt(((193150 cm⁴ - 102270 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 193150 cm⁴

Major principal second moment

[9] $I_2 = (I_y + I_z)/2 - \text{sqrt}(((I_y - I_z)/2)^2 + I_{yz}^2)$

= (193150 cm⁴ + 102270 cm⁴) / 2 - sqrt(((193150 cm⁴ - 102270 cm⁴) / 2)^2 + (0 cm⁴)^2)

= 102270 cm⁴

Minor principal second moment

[10] $\theta = ½\ \text{atan2}(-2\ I_{yz},\ I_y - I_z)$

= ½ atan2(0, 90880)

= 0 deg

Angle from the y-y axis to the major principal axis, anticlockwise

[11] $y_{top} = y_{max} - y_c$

= Green's theorem over 8 vertices in 2 loop(s)

= 300 mm

Extreme fibre above the centroid

[12] $y_{bot} = y_c - y_{min}$

= Green's theorem over 8 vertices in 2 loop(s)

= 300 mm

Extreme fibre below the centroid

[13] $x_{ext} = \max(x_{max} - x_c,\ x_c - x_{min})$

= Green's theorem over 8 vertices in 2 loop(s)

= 200 mm

Extreme fibre from the centroid, x

[14] $W_{el,y} = I_y\ /\ \max(y_{top},\ y_{bot})$

= 193150 cm⁴ / max(300 mm, 300 mm)

= 6438.4 cm³

Elastic section modulus about the y-y axis, governing fibre

[15] $W_{el,z} = I_z\ /\ x_{ext}$

= 102270 cm⁴ / 200 mm

= 5113.6 cm³

Elastic section modulus about the z-z axis

[16] $W_{pl,y} = |Q_{above}| + |Q_{below}|$ about the equal-area axis

= equal-area axis at y = 0 mm

= 7776 cm³

Plastic section modulus about the y-y axis

[17] $W_{pl,z} = |Q_{right}| + |Q_{left}|$ about the equal-area axis

= equal-area axis at x = 0 mm

= 5856 cm³

Plastic section modulus about the z-z axis

[18] $r_{g,y} = \text{sqrt}(I_y\ /\ A)$

= sqrt(193150 cm⁴ / 384 cm²)

= 224.28 mm

Radius of gyration about the y-y axis

[19] $r_{g,z} = \text{sqrt}(I_z\ /\ A)$

= sqrt(102270 cm⁴ / 384 cm²)

= 163.2 mm

Radius of gyration about the z-z axis

[20] $\alpha_{pl,y} = W_{pl,y} / W_{el,y}$
= 7776 cm³ / 6438.4 cm³
= 1.2078
Shape factor about the y-y axis

5. Results

SYMBOL	VALUE	DESCRIPTION
A	384 cm²	Area
I_y	193152 cm⁴	Second moment of area about the y-y axis
I_z	102272 cm⁴	Second moment of area about the z-z axis
I_{yz}	0 cm⁴	Product of inertia about the y-y and z-z axes
$W_{el,y}$	6438.4 cm³	Elastic section modulus about the y-y axis
$W_{el,z}$	5113.6 cm³	Elastic section modulus about the z-z axis
$W_{pl,y}$	7776 cm³	Plastic section modulus about the y-y axis
$W_{pl,z}$	5856 cm³	Plastic section modulus about the z-z axis
$r_{g,y}$	224.3 mm	Radius of gyration about the y-y axis
$r_{g,z}$	163.2 mm	Radius of gyration about the z-z axis
x_c	0 mm	Centroid x
y_c	0 mm	Centroid y
I_1	193152 cm⁴	Major principal second moment
I_2	102272 cm⁴	Minor principal second moment
θ	0°	Principal axis angle
$\alpha_{pl,y}$	1.2078	Shape factor about the y-y axis

6. Checks (governing first)

None.

7. Applicability

WARNING J: no torsion constant is computed for an arbitrary polygon. [no-torsion]

8. Verification

Template struct.section-properties@1.2.0 Content hash a95bd0059585bd36 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Seaburg PA, Carter CJ. Torsional Analysis of Structural Steel Members. Steel Design Guide 9. American Institute of Steel Construction; 1997.
3. British Standards Institution. Hot Rolled Steel Channels, I and H Sections—Dimensions and Masses. BS EN 10365. BSI.
4. British Standards Institution. Hot Finished Steel Structural Hollow Sections—Part 2: Tolerances, Dimensions and Sectional Properties. BS EN 10210-2. BSI.